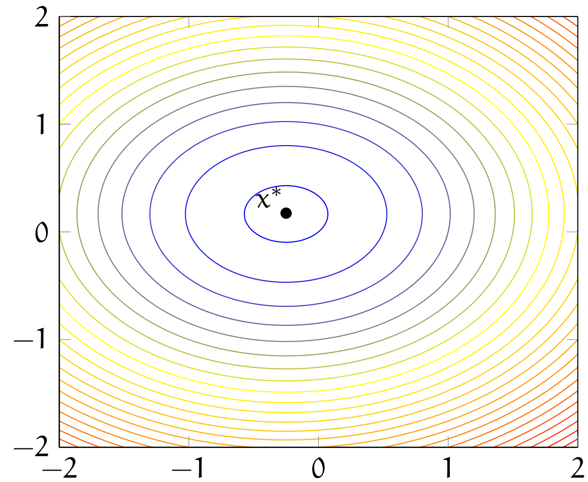
First, we show that the point

$$x^* = \begin{pmatrix} -\frac{1}{4} \\ \frac{1}{6} \end{pmatrix}.$$

is a local minimum of the function f. The gradient of the function is

$$\nabla f(x_1, x_2) = \begin{pmatrix} 4x_1 + 1 \\ 6x_2 - 1 \end{pmatrix},$$

which is zero at x^*.



The second derivative matrix is

$$\nabla^2 f(x_1, x_2) = \begin{pmatrix} 4 & 0 \\ 0 & 6 \end{pmatrix},$$

which is positive definite at x^*. Therefore, the point

$$x^* = \begin{pmatrix} -\frac{1}{4} \\ \frac{1}{6} \end{pmatrix}$$

satisfies the sufficient optimality conditions and is a local minimum.

The second derivative matrix is positive definite for all $x \in \mathbb{R}^2$, then f is strictly convex. Therefore, x^* is a unique global minimum of the function.

Thus, $x^* = (-\frac{1}{4}, \frac{1}{6})$ is the only global minimum of f.